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NATIONAL TECHNICAL UNIVERSITY OF UKRAINE “IGOR SIKORSKY KYIV
POLYTECHNIC INSTITUTE”
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INSTITUTE OF INFORMATION TECHNOLOGY AND LEARNING TOOLS OF THE NAES OF
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THE COUNCIL OF YOUNG SCIENTISTS OF THE FACULTY OF COMPUTER SCIENCE AND
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This book includes abstracts of the 8th International Conference "Information Technology and Implementation (Satellite) – 2021". Philosophical, theoretical and applied aspects which describe the results, problems and prospects of the creation and use of intelligent computing methods and creating of information systems and technology on their basis are reviewing.

Main tracks of the conference are: Artificial Intelligence Technologies, Cyberspace Protection Technologies, Data Analytics, Digital Project Management Technologies, E-commerce, E-government and E-learning Technologies, Mathematical Foundations of Information Technology, Network and Internet Technologies.

До цієї збірки увійшли тези 8-ї Міжнародної конференції «Інформаційні технології та впровадження (сателітної) – 2021». Розглядаються філософські, теоретичні та прикладні аспекти, які описують результати, проблеми та перспективи створення і використання інтелектуальних обчислювальних методів та створення на їх основі інформаційних систем і технологій.

Основні напрямки конференції: Технології штучного інтелекту, Технології захисту кіберпростору, Аналітика даних, Технології цифрового управління проєктами, Електронна комерція, Електронне урядування та Технології електронного навчання, Математичні основи інформаційних технологій, Мережеві та Інтернет-технології.

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¹ **Maryna Antonevych**

Master's student of the Faculty of Information Technology

² **Vitaliy Snytyuk**

Dr. of. Sci., Professor, Dean of the Faculty of Information Technology

^{1,2} *Taras Shevchenko National University of Kyiv*

DEVELOPMENT AND RESEARCH OF THE INTELLIGENT TECHNOLOGY FOR PREDICTING THE POPULARITY OF ANIMALS FROM THE SHELTER

Abstract. Millions of stray animals are injured on the streets every day around the world. So, what can be done, to help these animals to get new home? Scientists and researchers can suggest improvements that will help shelter animals to have a higher chance to be adopted with the help of data science. One of the solutions for this problem can be called one that will also use artificial intelligence. We intend to use a computer vision apparatus based on neural networks.

Keywords: Computer vision, prediction, intelligent technology, artificial intelligence, neural networks, deep learning, data science, shelter animals

Introduction. Millions of stray animals are injured on the streets every day around the world [1]. Such animals feel alone and they cannot live happy and healthy life. It is a great problem that so many animals cannot find their loving homes. In addition, these animals face many problems during their life in shelters. As example, for most dogs, being relinquished to an animal shelter is a drastic change and a very stressful experience [2]. Such environmental changes are very stressful for the most of dogs. Stress may also affect a shelter dog's physical health, causing increased susceptibility to diseases and a longer recovery time from illness [2].

And all of them need love and a cozy home. So, what can be done, to help these animals to get new home? One picture is worth a thousand words and the photography can save thousand lives. It is obvious that pets with charming photos can generate more interest and they can be adopted faster [1].

Scientists and researchers can determine the appeal of pet pictures. Moreover, they can suggest improvements that will help shelter animals to have a higher chance to be adopted with the help of data science. There are many animal shelters & organizations that help animals to get their new loving homes all over the word.

For example, "PetFinder.my" is Malaysia's leading animal welfare platform with over 180000 animals, 54000 of which have been safely adopted [1]. "PetFinder" works closely with animal lovers, the media, corporations and international organizations that help to improve animal welfare [1].

Moreover, they have launched Kaggle Competition on the 23rd of September, 2021. That's why it is new to the world. Their aim is to help shelter animals to get their new home with the help of data science.

Problem Formulation. As it Kaggle Competition, the given problem is posted on the site kaggle.com [1]. The task is to predict engagement with a pet's profile based on the photography for that profile. Therefore, our aim is to analyze raw images and

metadata and, after that, create intelligent technology that will be able to predict the “pawpularity” of pet photos.

Dataset. As it said in [1], the data can be used for academic research and education. In addition, the owners of the task have included optional Photo Metadata, manually labeling each photo for key visual quality and composition parameters. These labels are not used for deriving their “pawpularity” score, but it may be beneficial for better understanding the content and co-relating to a photo's attractiveness. The example of images is given in Fig.1.

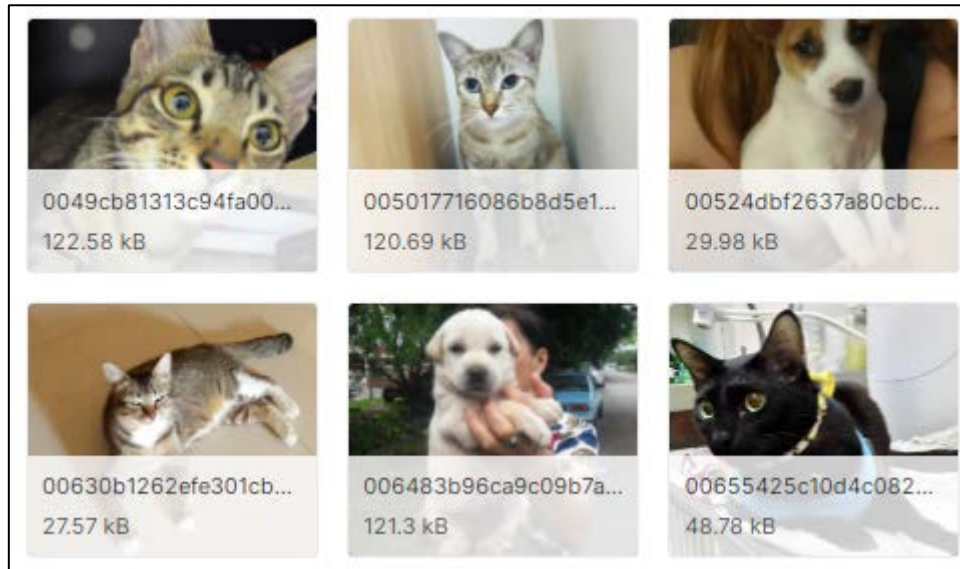


Figure 1 - Dataset

Previous Decisions Achieved. “PetFinder.my” currently uses a basic attraction counter for ranking pet photos. It analyzes image composition and other factors based on characteristics of thousands of pet profiles. The “pawpularity” score is derived from each pet profile's page view statistics at the listing pages, using the algorithm that normalizes the traffic data across different pages, platforms (web & mobile) and other various metrics. While this basic tool is useful, it is still in an experimental stage and the algorithm could be improved [1].

Offered Approach. One of the proposed solutions for this problem can be called the one that will use artificial intelligence (AI) technologies. As we have images in the dataset, so the solution belongs to the domain of the computer vision (CV) problems. CV is a field of AI that enables computers and systems to derive important information from digital images, videos and other visual inputs — and take actions or make recommendations based on that information. If AI enables computers to think, CV enables them to see, observe and understand [3].

For example, CV consists of such tasks as image processing, object detection, image recognition, image segmentation and so on.

Image processing is a method to perform some operations on an image, in order to get an enhanced image or to extract some useful information from it [4].

Object detection is a CV technique that allows to locate objects in an image or video. With this kind of identification and localization, object detection can be used to count objects in a scene and determine and track their precise locations, all while accurately labeling them [5].

Image recognition, in the context of machine vision, is the ability of software to identify objects in images [6].

Image segmentation is a method in which a digital image is broken down into various subgroups called “image segments” which helps in reducing the complexity of the image to make further processing or analysis of the image simpler. Segmentation in easy words is assigning labels to pixels. All picture elements or pixels belonging to the same category have a common label assigned to them [7].

All these tasks can be solved using such methods as: basic morphology operations (erosion, dilation, opening, closing etc.); machine learning methods (e.g. image segmentation can be done with the help of clustering methods [8], such as K-Means or evolutionary methods like genetic algorithms, MODS etc.); deep learning methods (e.g. Convolutional Neural Networks, Recurrent Neural Networks).

Therefore, we intend to use a computer vision apparatus based on neural networks for predicting the floating value of the “pawpularity” score.

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